

Portfolio Theory & Optimisation

UAPOML Summer Project – Week 1

Solutions: Questions 1–9

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Question 1

(a) **Simple Monthly Returns** $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$

- $R_1 = \frac{108-100}{100} = \frac{8}{100} = 0.0800$
- $R_2 = \frac{103-108}{108} = \frac{-5}{108} \approx -0.046296$
- $R_3 = \frac{115-103}{103} = \frac{12}{103} \approx 0.116505$
- $R_4 = \frac{110-115}{115} = \frac{-5}{115} \approx -0.043478$

(b) **Log Returns** $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

- $r_1 = \ln\left(\frac{108}{100}\right) = \ln(1.08) \approx 0.076961$
- $r_2 = \ln\left(\frac{103}{108}\right) = \ln(0.953703) \approx -0.047382$
- $r_3 = \ln\left(\frac{115}{103}\right) = \ln(1.116505) \approx 0.110261$
- $r_4 = \ln\left(\frac{110}{115}\right) = \ln(0.956522) \approx -0.044520$

(c) **Verification of the approximation** $r_t \approx R_t$ Using the Taylor expansion $\ln(1+x) \approx x - \frac{x^2}{2} + \dots$, the approximation error is $|R_t - r_t| \approx \frac{R_t^2}{2}$ so it grows with the magnitude of the return.

Month	R_t	r_t	$ R_t - r_t $	Comment
1	0.080000	0.076961	0.003039	moderate
2	-0.046296	-0.047382	0.001086	small
3	0.116505	0.110261	0.006244	largest
4	-0.043478	-0.044452	0.000974	smallest

The approximation is least accurate in Month 3, because $|R_3| \approx 11.65\%$ is the largest of the four returns in absolute value. Since the error scales roughly as $\frac{R_t^2}{2}$, larger returns produce larger approximation errors.

Question 2

(a) **Linearity of Expectation and $E[R_p]$** **Formal statement:** For any two random variables X, Y and constants $a, b \in \mathbb{R}$:

$$E[aX + bY] = aE[X] + bE[Y]$$

The property holds regardless of any dependence structure between X and Y .

Derivation: With $R_p = w_1R_1 + w_2R_2$, $w_1 = 0.4$, $w_2 = 0.6$:

$$E[R_p] = w_1E[R_1] + w_2E[R_2] = 0.4 \times 15\% + 0.6 \times 7\% = 6\% + 4.2\% = 10.2\%$$

(b) **Effect of Correlation on Expected Return** No, the result does not change. Linearity of expectation is a purely algebraic property of the expectation operator; it holds whether R_1 and R_2 are independent, positively correlated, or negatively correlated. Correlation affects variance (second moment) but never the mean (first moment).

Question 3

Claim:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$$

Proof:

Step 1. The portfolio return is $R_p = w_1R_1 + w_2R_2$. Hence, $\sigma_p^2 = \text{Var}(R_p) = \text{Var}(w_1R_1 + w_2R_2)$.

Step 2. Apply the given identity $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$ with $X = w_1R_1$ and $Y = w_2R_2$:

$$\text{Var}(w_1R_1 + w_2R_2) = \text{Var}(w_1R_1) + \text{Var}(w_2R_2) + 2\text{Cov}(w_1R_1, w_2R_2)$$

Step 3. Use the scaling properties of variance and covariance, $\text{Var}(cX) = c^2\text{Var}(X)$ and $\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$:

$$= w_1^2\text{Var}(R_1) + w_2^2\text{Var}(R_2) + 2w_1w_2\text{Cov}(R_1, R_2)$$

Step 4. Substitute $\text{Var}(R_i) = \sigma_i^2$ and the definition of the correlation coefficient, $\text{Cov}(R_1, R_2) = \rho_{12}\sigma_1\sigma_2$:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$$

Question 4

Parameters: $\mu_A = 18\%$, $\sigma_A = 25\%$, $\mu_B = 8\%$, $\sigma_B = 12\%$, $w_A = w_B = 0.5$.

The portfolio variance formula gives:

$$\begin{aligned}\sigma_p^2 &= (0.5)^2(0.25)^2 + (0.5)^2(0.12)^2 + 2(0.5)(0.5)\rho(0.25)(0.12) \\ &= 0.015625 + 0.0036 + 0.015\rho \\ &= 0.019225 + 0.015\rho\end{aligned}$$

(a) Portfolio Standard Deviations

- $\rho = +1$: $\sigma_p^2 = 0.019225 + 0.015 = 0.034225 \implies \sigma_p = \sqrt{0.034225} = 0.185 = 18.5\%$.
- $\rho = 0$: $\sigma_p^2 = 0.019225 \implies \sigma_p = \sqrt{0.019225} \approx 0.13865 = 13.87\%$.
- $\rho = -1$: $\sigma_p^2 = 0.019225 - 0.015 = 0.004225 \implies \sigma_p = \sqrt{0.004225} = 0.065 = 6.5\%$.

(b) When does σ_p equal the weighted average? $\sigma_p = w_A\sigma_A + w_B\sigma_B = 0.5 \times 0.25 + 0.5 \times 0.12 = 0.185$ when $\rho = +1$, i.e. 18.5%.

Algebraic proof: When $\rho_{12} = 1$:

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\sigma_A\sigma_B = (w_A\sigma_A + w_B\sigma_B)^2$$

Taking the positive square root: $\sigma_p = w_A\sigma_A + w_B\sigma_B$.

(c) Zero-Risk Weight when $\rho = -1$ When $\rho = -1$ the variance becomes:

$$\sigma_p^2 = (w_A\sigma_A - w_B\sigma_B)^2 = (w_A\sigma_A - (1 - w_A)\sigma_B)^2$$

Setting $\sigma_p^2 = 0$ requires $w_A^*\sigma_A = (1 - w_A^*)\sigma_B$ hence:

$$w_A = \frac{\sigma_B}{\sigma_A + \sigma_B} = \frac{0.12}{0.25 + 0.12} = \frac{12}{37} \approx 0.3243 \text{ (32.43\%)}. \quad \text{}$$

Question 5

From the two-asset portfolio variance formula,

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2$$

the final cross-term is strictly less than $2w_1w_2\sigma_1\sigma_2$ whenever $\rho_{12} < 1$. The weighted-average risk is $\bar{\sigma} = w_1\sigma_1 + w_2\sigma_2$ which satisfies $\bar{\sigma}^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2$ (the case $\rho_{12} = 1$).

Any reduction in ρ_{12} below 1 decreases the cross-term, so $\sigma_p^2 < \bar{\sigma}^2$ and thus $\sigma_p < \bar{\sigma}$. Economically, when assets do not move in perfect lockstep, losses in one asset are partially offset by gains in the other, cancelling out some total volatility.

Question 6

Parameters: $\sigma_1 = 0.25$, $\sigma_2 = 0.12$, $\sigma_3 = 0.04$, $\rho_{12} = 0.4$, $\rho_{13} = 0.1$, $\rho_{23} = 0.2$.

(a) **Covariance Matrix Σ** Using $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$ (and $\Sigma_{ii} = \sigma_i^2$):

- $\Sigma_{11} = (0.25)^2 = 0.062500$
- $\Sigma_{22} = (0.12)^2 = 0.014400$
- $\Sigma_{33} = (0.04)^2 = 0.001600$
- $\Sigma_{12} = \Sigma_{21} = 0.4 \times 0.25 \times 0.12 = 0.012000$
- $\Sigma_{13} = \Sigma_{31} = 0.1 \times 0.25 \times 0.04 = 0.001000$
- $\Sigma_{23} = \Sigma_{32} = 0.2 \times 0.12 \times 0.04 = 0.000960$

Therefore:

$$\Sigma = \begin{pmatrix} 0.062500 & 0.012000 & 0.001000 \\ 0.012000 & 0.014400 & 0.000960 \\ 0.001000 & 0.000960 & 0.001600 \end{pmatrix}$$

(b) **Symmetry Confirmation** By construction, $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j = \rho_{ji}\sigma_j\sigma_i = \Sigma_{ji}$ (since $\rho_{ij} = \rho_{ji}$). Inspection confirms $\Sigma_{12} = \Sigma_{21} = 0.012$, $\Sigma_{13} = \Sigma_{31} = 0.001$, and $\Sigma_{23} = \Sigma_{32} = 0.00096$. Hence $\Sigma = \Sigma^\top$.

Question 7

Risk-free rate $R_f = 4\%$. Sharpe Ratio: $S = \frac{E[R_p] - R_f}{\sigma_p}$.

(a) **Sharpe Ratio Calculations**

- $S_X = \frac{14\% - 4\%}{22\%} = \frac{10}{22} \approx 0.4545$
- $S_Y = \frac{9\% - 4\%}{10\%} = \frac{5}{10} = 0.5000$
- $S_Z = \frac{20\% - 4\%}{35\%} = \frac{16}{35} \approx 0.4571$

(b) **Ranking** $S_Y = 0.5000 > S_Z \approx 0.4571 > S_X \approx 0.4545$. Ranking: $Y > Z > X$.

A rational mean-variance investor prefers Portfolio Y because it delivers the highest excess return per unit of risk: for every 1% of volatility accepted, Y returns 0.50% above the risk-free rate, more than either X or Z. In the Capital Market Line framework, Y lies furthest north-west in risk-return space.

(c) **Why a Rational Investor May Prefer Y over Z** Portfolio Z has the highest absolute return (20%) but also the highest risk (35%). A rational mean-variance investor cares about risk-adjusted return. Z's Sharpe Ratio (0.457) is actually below Y's (0.500), meaning Z is relatively inefficient: the investor is not being adequately compensated for the extra volatility.

By holding Y and leveraging up (borrowing at R_f and investing in Y), the investor can replicate Z's expected return at lower risk, or achieve higher return at the same risk as Z. Hence, unless constrained, a rational investor prefers the portfolio with the higher Sharpe Ratio.

Question 8

(a) Global Minimum Variance (GMV) Portfolio The GMV portfolio is the portfolio of risky assets that achieves the smallest possible variance across all feasible weight vectors. Formally, it solves:

$$\min_w \text{ subject to } w^\top \mathbf{1} = 1$$

It is the leftmost point on the minimum-variance frontier because no combination of the same risky assets can produce a lower variance; any deviation in weights whether to increase or decrease expected return must increase volatility. Geometrically, the frontier curves rightward from the GMV in both directions.

(b) Should a Low-Risk-Tolerance Investor Hold a Portfolio on the Lower Half? No, this is incorrect. The minimum-variance frontier is a parabola in $(\sigma_p^2, E[R_p])$ space. Every portfolio on the lower half of the frontier is mean-variance dominated: for any such portfolio there exists another portfolio on the upper half with the same variance but strictly higher expected return. A rational investor regardless of risk tolerance should only ever hold portfolios on the efficient frontier (the upper half, from the GMV upwards), since these are not dominated. Even the most risk-averse investor should choose the GMV point itself, not a point below it.

(c) Construction and Slope of the Capital Market Line (CML) The CML is constructed by combining the risk-free asset (with return R_f and zero variance) with the tangency portfolio, the risky portfolio that maximises the Sharpe Ratio. Geometrically, it is the straight line drawn from the point $(0, R_f)$ on the expected-return axis tangent to the efficient frontier of risky assets. Every point on the CML corresponds to some linear combination of the risk-free asset and the tangency portfolio.

The slope of the CML is:

$$\text{slope} = \frac{E[R_T] - R_f}{\sigma_T}$$

where T denotes the tangency portfolio. This is precisely the Sharpe Ratio of the tangency portfolio—the maximum attainable Sharpe Ratio in the market—representing the additional expected return earned per unit of total risk along the efficient capital market line.

Question 9

(a) Standard Error of the Sample Mean with 36 Monthly Observations For a single asset with $\sigma_i = 30\%$ and $n = 36$ monthly observations, the standard error of the sample mean $\hat{\mu}_i$ is:

$$SE(\hat{\mu}_i) = \frac{\sigma_i}{\sqrt{n}} = \frac{0.30}{\sqrt{36}} = \frac{0.30}{6} = 0.05 = 5\%$$

This is an extremely large estimation uncertainty: the true expected return could plausibly lie anywhere in a $\pm 10\%$ range around the estimate (a two-standard-error band). It implies that historical sample means are too noisy to be used directly as inputs to MVO; small differences in estimated μ will induce large, unstable swings in optimal portfolio weights.

(b) Error Maximisation and the Role of ML-Predicted Returns **Error maximisation:** MVO systematically over-weights assets with over-estimated expected returns and under-weights those with under-estimated returns. Because the optimiser exploits apparent return differences without knowing which are genuine and which are estimation noise, it amplifies errors in μ directly into extreme and unstable portfolio weights. This is known as error maximisation.

How ML-predicted returns help: Machine learning models (e.g., gradient-boosted trees or neural networks) incorporate a rich set of predictive features (momentum signals, fundamental ratios, macro factors) and use regularisation techniques that shrink noisy estimates toward more sensible values. The resulting $\hat{\mu}^{ML}$ has lower out-of-sample variance than a raw historical mean, reducing the magnitude of estimation errors fed into the optimiser and thereby yielding more stable, better-diversified portfolios.

(c) Number of Unique Parameters in Σ for $N = 50$ Assets The covariance matrix Σ is symmetric. The number of unique parameters is:

$$\underbrace{N}_{\text{variances}} + \underbrace{\frac{N(N-1)}{2}}_{\text{unique covariances}} = 50 + 1225 = 1275$$

This is a major estimation challenge because with, say, 3 years of monthly data ($n = 36$ observations) there are far more parameters to estimate than observations ($1275 \gg 36$). The resulting sample covariance matrix is ill-conditioned (often singular or nearly so), producing unreliable portfolio weights. Practitioners address this via shrinkage estimators, factor models, or Bayesian methods that impose structure on Σ .